Practice Assignment

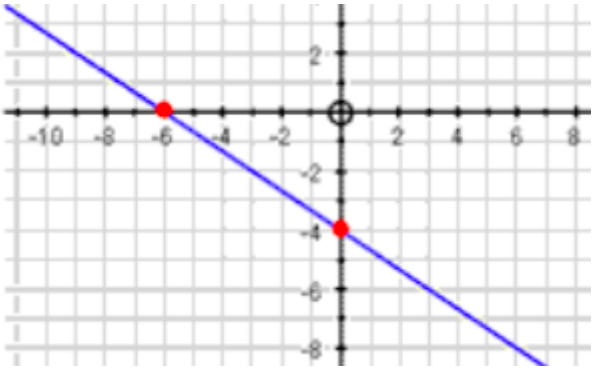
1. **Simplifying Expressions:** Simplify the expression $3(2x+4)-5x^3(2x + 4) - 5x^3(2x+4)-5x$.
2. **Solving Equations:** Solve the equation $2x-3=7$.

3. **Factoring:** Factor the quadratic expression x^2-5x+6.
4. **Quadratic Formula:** Use the quadratic formula to find the roots of the equation $x^2+4x-5=0$

5. **Evaluating Functions:** If $f(x)=2x^2 - 1$ what is $f(-4)$?

7. **Word Problem:** A rectangle has a length that is 3 meters more than its width. If the perimeter is 30 meters, what are the dimensions of the rectangle?
8. **Interest Problem:** If you invest $500 at an interest rate of 5% per year, how much money will you have after 3 years if the interest is compounded annually?

9. **Graphing Linear Equations:** What is the slope and y-intercept of the equation $y=-2x+5$?
10. Find the largest factor of the number 818346 which is less than the number itself.
11. Find the magnitude(modulus) and phase(angle) of a complex number (-3+3i).
12. Write down the equation of the straight line shown in Fig. 1 below.



13. Consider a prime number 1394849. Find the largest factor of the number which is smaller than the number itself.

14. Find 10th term in the sequence 4, 12, 36, 108, , , ,